

From link position to the physics of motion

M9 • 03 Jun 2027 - 06 Jul 2027 • 75 hours

What you will produce by the end of the month

The deliverable for this month is a joint model using the parameters identified in M7, together with a dynamics simulation of a two-link mechanism or leg. Check dimensions, the energy balance and agreement between calculations and measurements. The kinematics from M8 finds the angles that place a link endpoint at a specified point; dynamics relates motion to the required torque and actuator limits. For a leg, it describes the effects of link masses, body motion and foot contact; for a finger, it describes phalange acceleration, load holding and the reaction when touching an object. Begin with one link, then add a second link and contact.

Inputs

Bring the geometry, coordinate frames, FK and Jacobians from M8. From M7, bring angle, command and current measurements and the available actuator model, including friction, delay and the limits of identification. From M5, bring CAD-part masses and centres of mass, provided they were determined correctly. Label each quantity as measured, calculated in CAD, taken from manufacturer documentation or specified for an illustrative example. Inertia calculated in CAD using a default density is not the measured inertia of a printed part. Account for fasteners, wiring and actual infill; if data are unavailable, state the uncertainty of the estimate.

Initial self-check: distinguish kg from N, N from N·m, velocity from acceleration, and inertia about the centre of mass from inertia about the joint axis. Check the work done by a force through a small displacement and calculate the energy of a load at a known height. Make sure you know the axis from which q is measured and the direction of gravity. An incorrect potential-energy sign can produce smooth simulated motion that contradicts physics.

Scope and evidence

M9 does not require the dynamics of a complete multi-legged platform with gait control. The basic model has a fixed base, two links and explicitly selected contacts. Increase complexity only after checking simple limiting cases. Simulation can confirm the correctness of the equations and numerical method, but agreement with a real actuator requires comparison against a separate measured run. If M7 did not produce physical measurements, complete the computational work and mark the comparison with the real actuator as unfinished.

Prepare a working integrator and a description of the model's limits of applicability. State the angle, velocity and load ranges and contact modes that you have checked. Separately describe the conditions under which backlash, saturation, compliance, heating or an unknown internal actuator controller become significant. Subsequent modules will use this information to choose control laws that account for model accuracy.

75 hours: from simple models to complex ones

Four blocks

Hours 1-20: Newton's and Euler's equations, mass, centre of mass, inertia, forces and energy; 10 hours of theory and 10 hours of practice. Hours 21-40: Lagrange's equations, equation structure and two-link dynamics; 10+10. Hours 41-60: normal contact, friction, impact and static support; 10+10. Hours 61-75: use of the M7 identification results, model validation, sensitivity and error assessment; 5+10. This totals 35 hours of study and analysis and 40 hours of implementation and experiments. The allocation includes documenting results rather than leaving the report to uncounted time.

Divide each of the first three 10-hour theory blocks into 2-hour sessions: reading a section, deriving equations by hand, checking units and preparing an experiment. Schedule the 10-hour practical blocks on Sundays, allowing breaks between stages. Split the final 5 hours of theory as 2+2+1. The module dates establish its place in the overall calendar; retain the boundary dates, rest days, holidays and separate integration checks. Hour numbers express the workload: do not count the same time twice when placing sessions in the calendar.

The dynamics-lab project structure

parameters/ contains masses, lengths, centres of mass, inertias and their provenance. models/ contains the single-link, 2R-mechanism and contact models. experiments/ stores inputs, initial states and integration settings. data/measured/ preserves the M7 data unchanged; data/simulated/ stores computational results. validation/ contains comparison results and runs for independent validation; reports/ contains energy balances and model limits. Each simulation records the equation version, solver step size or tolerances, duration, input signals and termination reason.

When to increase complexity

First, check the energy of one link without friction or external input. Second, add a known torque and compare its work with the energy change. Third, introduce viscous friction and check that energy decreases. Fourth, repeat these checks for two links. Fifth, add contact and account for energy losses. Sixth, use the real actuator model from M7 and an independent measurement recording. If a basic check fails, find the cause first; do not try to remove the discrepancy by increasing damping or fitting a physical parameter.

Separate measurement uncertainty from numerical error. Changing mass explores sensitivity to a physical parameter; reducing the integrator step explores numerical error. Include both experiments in the report; one cannot replace the other. At the outset, choose several simple motions that you can explain without a computer, and revisit them after adding each new equation term.

Newton and Euler: forces, torques and inertia

How to draw a force and torque diagram

For translation of the centre of mass, the sum of external forces equals $m \cdot a_{\text{COM}}$. For planar rotation of a rigid link about a fixed axis, sum torques about that axis and use the moment of inertia about the same axis. In general spatial motion, the rotational equation includes $I \cdot \dot{\omega}$ and the gyroscopic term $\omega \times (I\omega)$, expressed in a consistent basis. Do not automatically apply the planar relation $\tau = I\alpha$ to an arbitrary rotating system. The centre of mass is the point whose motion describes the translational part of the body's motion; the mass distribution relative to the axes determines its inertia.

Task 1. Physical parameters of a link

The inputs are the M5 CAD link and available mass measurements. 1. Separate geometry from material and list the contributions of fasteners and the actuator. 2. Determine mass, COM position in the local coordinate frame and inertia about the COM. 3. Transfer the inertia to the joint axis using the parallel-axis theorem; in the planar case, $I_{\text{joint}} = I_{\text{COM}} + m \cdot d^2$. 4. Compare with a rod or point-mass approximation. Deliverable: a parameter table with units and sources. Check: shifting the axis away from the COM increases inertia by a non-negative amount. Edge case: an incorrect CAD scale changes inertia more strongly than it changes a linear dimension.

Task 2. Forces and torques on a leg or finger link

Draw one link showing its weight, joint reaction force, actuator torque and an external force at its endpoint. 1. Choose the point about which you calculate torques. 2. Write down lever arms and signs. 3. Calculate the holding torque in several known configurations. 4. Check the external-force torque using $\tau = J^T F$ from M8; calculate the gravitational contribution separately. Deliverable: three labelled drawings and a torque table. Check: if a force's line of action passes through the axis, its torque about that axis is zero. Error to avoid: a link's mass cannot be added as a force without multiplying by g .

Energy and units

A single link's kinetic energy is the sum of $0.5m \cdot v_{\text{COM}}^2$ and $0.5\omega^T I_{\text{COM}} \omega$. Potential energy in a uniform gravitational field depends on COM height. For two links, add their energies; obtain COM velocities from the geometry and Jacobians in M8. Prepare a table: length m , mass kg , time s , force N , torque $N \cdot m$, energy J , inertia $kg \cdot m^2$. Check the dimensions of every term before running a numerical calculation. Using I for both inertia and electric current in different files is acceptable only when the context is clear; separate names such as inertia and current are preferable.

One link: its equation and an energy test

Task 3. A single-joint pendulum

For clarity, measure θ from the downward vertical. With moment of inertia J about the axis, mass m and distance c to the COM, a simple model is $J\ddot{\theta} + b\dot{\theta} + m\cdot g\cdot c\cdot \sin(\theta) = \tau$. A convenient zero for potential energy is the lowest point: $V = m\cdot g\cdot c\cdot (1 - \cos\theta)$. If you use an angle q measured from the horizontal axis, the gravitational term changes; do not switch conventions without transforming the coordinates. 1. Derive the gravitational torque from a drawing. 2. Write the state $[\theta, \omega]$ and its derivative $[\omega, \alpha]$. 3. Solve the equation for α , then integrate. 4. Compare the initial acceleration direction with a hand calculation.

Deliverable: angle, velocity and energy plots, plus a configuration that reproduces the calculation. Check without friction or torque: total energy should be conserved within numerical error, which you investigate by reducing the step size or tolerances. For a small angle, compare the period with $2\pi\sqrt{J/(m\cdot g\cdot c)}$ for positive parameters. Edge cases: the lowest point with zero initial velocity remains an equilibrium; reject negative mass or zero inertia as an invalid model.

Task 4. Work and dissipation

First disable gravity and apply a constant torque to a link with known inertia J : acceleration should be τ/J . Then restore gravity and apply a bounded torque pulse. Numerically integrate power $P = \tau\cdot\omega$ and compare work with the energy change. Add viscous friction $b\cdot\omega$ with $b \geq 0$: the dissipated power $b\cdot\omega^2$ is non-negative. Save the balance $E(t) - E(0) - \text{actuator_work} + \text{losses}$, as well as the q plot. Check: with zero actuator torque and positive damping, mechanical energy must not systematically increase. Negative b is a separate artificial test that the validator must reject in a passive model.

The numerical method is part of the experiment

Compare fixed-step integration with the adaptive `solve_ivp` solver. Record duration, step size, relative and absolute tolerances, and termination status. Run an experiment with successive step-size reductions while keeping physical parameters unchanged. Assess both the trajectory and the energy-balance residual. Similar position plots do not yet establish convergence of force or acceleration. High velocities and stiff contact will change the integrator requirements; preserve this basic model without contact for subsequent comparisons.

Lagrange's equations and the structure of two-link dynamics

What to derive

The generalised coordinates $q=[q_1, q_2]$ uniquely specify the 2R configuration. Measure q_1 from the horizontal x-axis and q_2 relative to the first link; use the x-y plane with y upward. Construct the centre-of-mass positions, differentiate them with respect to time, and calculate kinetic energy K and potential energy V . The Lagrangian is $L=K-V$. For each q_i , write $d/dt(\partial L/\partial \dot{q}_i) - \partial L/\partial q_i = \tau_i$, adding external forces and losses separately. A symbolic algebra system can help with derivatives, but you must be able to explain every energy expression and the choice of signs.

Task 5. Mass matrix and gravity

Use l_1, l_2 for lengths, c_1, c_2 for distances from the corresponding joints to the COMs, m_1, m_2 for masses, and I_1, I_2 for planar inertias about the COMs. Extract the structure $M(q) \cdot \ddot{q} + h(q, \dot{q}) + g(q) = \tau$ from the equations. For checking: $M_{22} = I_2 + m_2 \cdot c_2^2$, $M_{12} = M_{21} = I_2 + m_2 \cdot (c_2^2 + l_1 \cdot c_2 \cdot \cos q_2)$, $M_{11} = I_1 + I_2 + m_1 \cdot c_1^2 + m_2 \cdot (l_1^2 + c_2^2 + 2l_1 \cdot c_2 \cdot \cos q_2)$. Here h combines the velocity-dependent terms; an individual matrix C is not unique, so check the vector h first.

1. Obtain M and g from your own derivation. 2. Compare M against an independent implementation or the structure given above. 3. Check symmetry and positive definiteness in admissible configurations. 4. Compare gravitational torque with the numerical derivative of V with respect to q . Deliverable: a computational notebook with the derivation, a numerical function and tests. For the chosen coordinates, $g_2 = m_2 \cdot g \cdot c_2 \cdot \cos(q_1 + q_2)$, while g_1 additionally contains $g \cdot (m_1 \cdot c_1 + m_2 \cdot l_1) \cdot \cos q_1$. Edge case: all h terms must vanish when $\dot{q} = 0$.

Task 6. Forward and inverse dynamics

The inverse problem takes $q, \dot{q}, \ddot{q}$ and returns the required τ ; the forward problem takes $q, \dot{q}, \tau$ and solves a linear system for $\ddot{q}$. 1. Choose several states. 2. Calculate τ using inverse dynamics. 3. Feed it into forward dynamics and compare the reconstructed acceleration. 4. Repeat with endpoint forces added through $J^T F$ in a consistent basis. Save the acceleration reconstruction errors after the inverse-then-forward calculation. Use $\text{solve}(M, \text{rhs})$, not an explicit matrix inverse. This test does not replace an independent physical check: both functions may contain the same model error.

2R simulation and checks of physical behaviour

Task 7. Two links without contact

Inputs: validated M, h, g , the M8 geometry, and physically admissible masses and inertias. 1. Run free motion from a non-zero initial configuration without friction. 2. Plot angles, velocities, the endpoint path and total energy. 3. Repeat with half the step size or tighter tolerances. 4. Add damping and bounded torque, checking the work balance. Deliverable: four comparable runs and a numerical-convergence report. Edge case: with horizontal links and no holding torque, acceleration must not be zero merely because the initial velocities are zero.

Checking velocity-dependent terms

For the 2R model described above, define $a = m_2 \cdot l_1 \cdot c_2 \cdot \sin(q_2)$. Then $h_1 = -a \cdot (2\dot{q}_1\dot{q}_2 + \dot{q}_2^2)$, $h_2 = a \cdot \dot{q}_1^2$. Compare this with your own derivation and check reversal of velocity direction. Do not remove h simply because calculations agree with an experiment at low velocity: the error will change as velocities change. However, a slow-motion model may deliberately neglect some dynamics if this is an explicit approximation and its error has been checked. Record the velocity range in which the approximation remains useful.

Application to a leg and hand

Use the 2R model as a planar section of a leg or as two phalanges. For the leg, add the masses of the relevant parts and actuators mounted on the links; for the finger, specify a different length scale and fingertip load. Compare gravitational torque and inertial effects for the same normalised motion. Do not transfer planar-model equations to a complete spatial leg without a new derivation: the first axis, COM offsets and inertia tensors change the equations. It is sufficient to specify the interface through which a multibody library will later connect.

An independent numerical reference

Build the same simple mechanism in one tool, such as Drake, MuJoCo or Featherstone's reference library. First disable contact and default actuators, and align gravity, masses, COMs, inertias and initial coordinates. Compare acceleration in one pose before comparing a long trajectory; then compare energies and a short motion. Disagreement may result from built-in damping, limit stops or a different coordinate frame. In the comparison report, list the effects that are enabled and disabled. You need not install several heavyweight simulators for one plot: one independent reference is enough.

Contact, friction and body support

Task 8. Unilateral normal contact

A horizontal surface can exert an upward force on a foot, but cannot attract it towards the floor. Begin with a vertical mass and a spring-damper model of penetration $\delta \geq 0$. An illustrative model force $N = \max(0, k \cdot \delta + c \cdot \dot{\delta})$ acts only during contact; δ is positive when penetration increases. 1. Check free fall before touchdown. 2. Add contact and measure maximum penetration. 3. Check the force sign and absence of tension during lift-off. 4. Compare several k, c values and integrator step sizes. Deliverable: height, force, energy and the region of numerical stability. These parameters do not automatically represent the properties of a real foot.

Task 9. Sticking and sliding

In a simple Coulomb description, static tangential force is bounded by $|F_t| \leq \mu_s N$; during sliding it opposes velocity, with a magnitude determined by $\mu_k N$. 1. Place a body on a horizontal plane. 2. Gradually increase the prescribed horizontal force. 3. Show the transition from sticking to sliding. 4. Repeat with a different normal load. Deliverable: plots of tangential force, velocity and contact mode. Check: tangential friction must not systematically add mechanical energy. Edge case $N=0$: the contact friction force also vanishes. Smoothing near zero velocity changes the model and must be explicitly identified.

Impact and discrete time

Compare elastic and inelastic impact with a continuous compliant model. The coefficient of restitution describes the ratio of normal velocities after and before impact under specified conditions; it is not a universal material coefficient for every velocity. Do not combine a rigid impulsive impact with the same energy loss in a damper without checking for double counting. For the first implementation, explain the two approaches and implement one, then investigate sensitivity. Force, penetration and energy matter in contact; smooth animation can hide non-physical high-frequency oscillations.

Task 10. Support polygon

Take known contact points on a horizontal plane and the projection of the body COM. Construct the convex hull of active contacts and the signed distance from the projection to its boundary: positive inside and negative outside. Repeat after removing one support and shifting the COM. Deliverable: support diagrams and the static margin in metres. Check: a point on an edge has zero margin; two contacts do not form a polygon with non-zero area. This is a quasi-static geometric check under stated force and friction assumptions, not proof of the dynamic stability of a walking robot.

Validating the M7 model and assessing its accuracy

Task 11. A joint with real actuator parameters

First establish what M7 identified: mechanics under a known torque, or the response of a servo with an internal controller to a position command. 1. Import parameters with units, signs and operating ranges. 2. Check gearing and inertia: do not count the load twice. 3. Replay the measured input in simulation with the same initial state. 4. Compare $q(t)$ and the available velocities and currents on a common time axis. Deliverable: overlaid plots, a residual and an explanation of the most prominent discrepancy. Do not connect a position-command response model as an ideal torque source: these are different physical interfaces.

If torque is not directly known, validate the available command-to-angle model and state separately which mechanical properties remain unconfirmed. Current telemetry can support an explanation of saturation, but does not become an accurate torque measurement without a coefficient and a definition of the current type. Add heating, friction, backlash and delay from M7 only where their meaning is consistent with the new equation. For each effect, first show a comparison against the model without it; otherwise you cannot determine what improved the result.

Task 12. An independent recording and sensitivity

Before fitting parameters, set aside recordings of separate runs for independent validation. 1. Fit the model on the first group. 2. Freeze the parameters. 3. Predict a new run with a different admissible command waveform or load. 4. Calculate angle RMSE, maximum residual and characteristic-time error. Then vary one parameter at a time within its justified uncertainty range and compare the effects. Deliverable: a sensitivity table and independent-validation plots. If no new real data are available, perform a software test, but leave physical validation open.

Assessing model errors

Prepare a table of error source, observed effect, required accuracy and refinement method. For example: COM affects gravitational torque; J affects acceleration; delay shifts the response; friction affects low velocities; contact k affects penetration and peak force. Allocate time to the parameters that most strongly affect the result of interest. State numerical integration error and measurement noise separately. The aim is the smallest model adequate for the chosen experiment, rather than the largest number of parameters. The report must identify admissible operating conditions and specific examples where the model ceases to be reliable.

Step by step: hours 1-40

Hours 1-10: physical terminology and diagrams

1-2: check units, gravity and the coordinate frame, and assemble the parameter table from M5-M8. 3-4: study mass, COM and inertia, and transfer inertia to a joint axis for a simple body. 5-6: draw force and torque diagrams for a link and leg, and calculate static torques. 7-8: work through kinetic and potential energy and the work done by torque. 9-10: derive the pendulum equation for the chosen angle and define states and initial conditions. These ten hours do not require a complex simulation: their outputs are checkable equations and physically expected relationships.

Hours 11-20: one link as a test system

11-12: finish the parameter table for task 1 and check the provenance of CAD mass and inertia. 13-14: complete the drawings and calculations for task 2 and compare against JTF. 15-16: implement the pendulum in task 3 and plot q, ω, E . 17-18: complete the energy balance in task 4 with torque and friction. 19-20: investigate the integrator step size and compare small oscillations against the analytical period. Freeze baseline-one-link with its inputs, method and achieved accuracy. If energy grows without a source, fix the model before moving to the second link.

Hours 21-30: the two-link system

21-22: write the 2R centre-of-mass coordinates and their velocities using the M8 geometry. 23-24: construct K, V, L and derive Lagrange's equations. 25-26: extract the mass matrix, velocity-dependent terms and gravity. 27-28: study the meaning of forward and inverse dynamics and work through adding an external force. 29-30: prepare checks of M symmetry, positive energy, the derivative of V and limiting cases. If using a symbolic tool, preserve the original energy expressions and the meanings of variables, as well as the final long formula.

Hours 31-40: validated 2R dynamics

31-32: implement M, h, g for task 5 and compare with your own derivation. 33-34: check the inverse-then-forward dynamics calculation in task 6 using fixed and random admissible states. 35-36: run the contact-free simulation in task 7 and its energy checks. 37-38: add bounded torque and friction, and repeat the convergence check. 39-40: compare instantaneous acceleration and a short trajectory with one independent library. If installation is taking too long, use an available reference implementation; do not reduce unit and energy checks to accommodate a simulator interface.

Step by step: hours 41-75

Hours 41-50: contact physics

41-42: study unilateral normal force, penetration, and the difference between rigid and compliant contact. 43-44: work through the friction cone and the transition from sticking to sliding, and draw admissible forces. 45-46: study impact, the coefficient of restitution and energy losses. 47-48: assess the effects of contact stiffness, damping and integration step separately; prepare a parameter grid for comparison. 49-50: study the support polygon and static margin, and record their conditions of applicability. Relate every concept to an observable force, velocity or energy.

Hours 51-60: contact experiments

51-52: implement the vertical contact in task 8; check the absence of force before touchdown and at lift-off. 53-54: vary k, c and the numerical step, recording penetration and energy. 55-56: complete the sticking-to-sliding transition in task 9 and check the sign of frictional power. 57-58: construct several support configurations for task 10, remove a contact and move the COM. 59-60: add simple endpoint contact to the 2R mechanism or leg in the chosen simulator and explain how it differs from a geometric check. If time is limited, retain point contact, while preserving the unilaterality and sensitivity checks.

Hours 61-65: model validation plan

61-62: revisit M7 identification and align the input type, delay, gearing and load inertia with the M9 model. 63-64: select separate measured runs and metrics, estimate errors and list parameters for sensitivity analysis. 65: finalise the table of validity regions and the final report format. These 5 hours are split into 2+2+1 without adding time to the original programme. Do not revise the quality criterion after seeing validation results in a way that makes every outcome a success.

Hours 66-75: compare calculation with experiment

66-67: complete the transfer in task 11 and reproduce the measured input and initial conditions. 68-69: compare trajectories, fix unit, timing and interface errors, then freeze the parameters. 70-71: perform the independent validation in task 12 and calculate errors using the predefined metrics. 72-73: conduct sensitivity analysis and separate numerical problems from physical disagreement. 74-75: assemble supporting results, limitations, a reproducible run and a brief analysis of one failed model. If time is short, reduce contact-geometry complexity and additional tools; the identified joint, 2R or leg dynamics, energy checks and independent model validation remain mandatory.

Acceptance checks: detecting physical errors

Test scenarios

F1: no gravity, friction or external torque. Rest is maintained; with an initial velocity, the corresponding energy balance holds. F2: one link under constant torque without gravity. Acceleration agrees with τ/J . F3: positive viscous friction without actuation. Energy decreases within integration error. F4: geometry is scaled incorrectly. The unit table and comparison against the analytical period reveal the error; visual similarity of the trajectory alone is insufficient to accept the simulation. F5: the mass matrix is asymmetric or not positive definite for a physical configuration. The test rejects the model before integration.

F6: contact without penetration and during lift-off. No attractive normal force occurs. F7: tangential sliding. Frictional work does not create energy; contact force vanishes when $N=0$. F8: integration step size is reduced without changing physical parameters. The result must show convergence of the metrics of interest; otherwise conclusions about mechanics are premature. F9: the COM moves outside the horizontal static support polygon. The margin changes sign; the test does not automatically conclude that dynamic falling occurs. F10: parameters are fitted to one run, but do not reproduce a new one. Narrow the validity region or improve the model and repeat independent validation.

Evidence that the module is complete

Provide dimensional and energy tests; checks of M, h, g ; input parameters with provenance; a two-link or leg simulation; contact sensitivity analysis; transfer of the M7 identification results; and comparison of a measured trend with calculation. The measured trend may be delay, characteristic time, load dependence or decay; choose the metric in advance. The report must explicitly identify conditions under which the model fails. Without a physical recording, the corresponding item remains open even if every mathematical test passes.

Questions and handover to the next module

Explain why moment of inertia depends on the axis; where velocity-dependent terms come from; how to check the gravity sign; why to solve a system with M rather than calculate its inverse; why excessively stiff contact complicates integration; how the friction cone differs from the static support-polygon rule; why a servo model with an internal controller cannot be treated as an ideal torque actuator; and how to distinguish noise, numerical error and incorrect physics. Pass on parameters, original measurements, dynamics functions and accuracy limitations. When designing the controller next month, account for model validity limits and actuator saturation; calling a simulation function alone is insufficient.

Assigned reading: a focused reading route

Books and lecture notes may be in English; keep the plan, definitions and your own explanations in English. Read only the specified sections within the existing 75 hours.

Core reading and one reference route

1. Core: K. Lynch, F. Park, Modern Robotics, Chapter 8. §8.1.3 - the mass matrix; §8.3 - Newton-Euler; §8.5 - forward dynamics. Tasks 1-7 require force diagrams, the meaning of $M(q)$ and an independent check of the forward and inverse problems.

2. Core: R. Tedrake, Underactuated Robotics. The Simple Pendulum: Equations of Motion; Multi-Body Dynamics: Lagrangian Dynamics and Manipulator Equations; Planning and Control through Contact: physical models of contact and friction. Read for tasks 3-10, without trajectory optimisation.

3. Reference: R. Featherstone, Rigid Body Dynamics Algorithms. Chapters Modelling Rigid Body Systems and Inverse Dynamics: establish the definitions of inertia and external force used, and the directions of recursive calculations. The relevant example is sufficient for checking task 6; the complete set of algorithms is not yet required.

Documentation for specific practical work

SciPy solve_ivp - method, rtol and atol parameters, the events mechanism and the success flag for tasks 3-7. Featherstone spatial_v2 - mcl, HandC, ID and FDab as reference implementations of dynamics.

MuJoCo Computation - Contact and Numerical Integration if comparing simulators; installing MuJoCo is not required for your own numerical experiment.

After reading

After the derivation, check units, the gravity sign and the energy balance. Compare two integration step sizes, then change one physical parameter. For tasks 11-12, return to the actuator model and validation data from M7. The presence of contact alone does not imply the absence of sliding.

Key terms and definitions

Rigid body. An idealisation in which the distances between any two points of a body remain constant during motion and loading.

Centre of mass. The point whose position is the average position of all body elements, weighted by their masses.

Inertia tensor. A matrix describing mass distribution about a selected point; it determines the rotational contribution to kinetic energy and has units $\text{kg}\cdot\text{m}^2$.

Free-body diagram. A diagram of an isolated body showing all external forces and torques, including reactions from removed constraints.

Generalised coordinates. Independent configuration parameters sufficient to describe the system's position under its geometric constraints.

Lagrangian. The difference between kinetic and potential energy, used in Lagrange's equations to obtain system dynamics.

Mass matrix. The configuration-dependent matrix $M(q)$, which determines kinetic energy and relates generalised accelerations to forces.

Velocity-dependent terms. The part of the dynamics arising from coordinate motion, including Coriolis and centrifugal effects; the representation of matrix C is not unique.

Gravity vector. Generalised torques or forces corresponding to the potential-energy gradient under the chosen equation-of-motion convention.

Forward dynamics. Calculation of accelerations from current coordinates, velocities, applied forces and the chosen constraint model.

Inverse dynamics. Calculation of required generalised forces from specified coordinates, velocities and accelerations with known external inputs.

Dissipation. Irreversible conversion of mechanical energy into other forms, such as heat, through friction or damping.

Unilateral contact. A constraint that permits bodies to separate and contact to be compressed, but does not permit an attractive normal force between surfaces.

Normal force. The component of contact force along the surface normal; in a model without adhesion it points away from the surface.

Friction cone. The set of admissible contact forces in the Coulomb model: tangential force magnitude is bounded by μ times the normal force.

Sticking. A contact mode with zero relative tangential velocity, possible while the required force remains within the friction limit.

Impulse. The time integral of force, which changes momentum; it approximates a short impact without detailed calculation of peak force.

Support polygon. The convex hull of support points on a plane; the centre-of-mass projection provides only a quasi-static criterion.

Check the relationships: $M(q)\ddot{q} + c(q, \dot{q}) + g(q) = \tau + J^T f$. All forces must have consistent signs, bases and points of application.

Preparation for dynamics experiments

NEW PURCHASES: NONE REQUIRED

Use the existing MacBook Pro M4, CAD link models from M5, IMU and controller from M6, and the secured actuator and current and temperature measurement equipment from M7. Investigate 2R dynamics and contact numerically. The 3D printer is needed only if the fixtures change; choose a printing profile for the exact printer model.

PREPARE BEFORE HOURS 1-20

Create a parameter table for each link: mass in kg, centre-of-mass position in m, inertia in $\text{kg}\cdot\text{m}^2$, coordinate frame and value source. Distinguish inertia about the centre of mass from inertia about the joint axis. Mark estimated parameters and their uncertainty ranges.

Before repeating a physical experiment, check mounting, stopping, the M7 limits, angle direction and consistency of measurement timestamps. Use loads and levers from the existing fixtures; increase the load only within the previously justified operating range.

INSTALL NOW

No new packages are mandatory. On macOS ARM, use the Python, NumPy, SciPy, Matplotlib and pytest installations from M8; use SymPy if you already selected it for your own symbolic derivation. [SciPy solve_ivp](#) integrates your own equation system on the CPU.

OPTIONAL: INDEPENDENT COMPARISON

An external engine is needed only for the chosen independent comparison. This month, you can compare equations and values with [the author's spatial_v2 library](#) without installing MATLAB/Simulink. MuJoCo and other engines are not new mandatory purchases or prerequisites for completing M9.

CHECK THE ENVIRONMENT AND PHYSICAL MODEL

Reproduce pendulum motion: without friction, check energy drift as the step size decreases; with friction, check that energy decreases. For 2R, check symmetry and positive definiteness of M , and consistency between forward and inverse dynamics. In the contact model, check the absence of attraction to the surface and the distinction between sticking and sliding.

Save versions, integrator configuration, initial conditions, an energy plot and an independent validation recording. A successful simulation confirms only the model properties tested; physical robot readiness requires its own experiments. Preparation is included in the existing 75 hours.

Motion fundamentals: why torque is needed

Required: derive the equation of motion in two ways

The kinematics in M8 relates angles to endpoint position. Dynamics determines which torque produces a chosen acceleration and what happens with a limited actuator. Before deriving the equations, clarify the distinction between mass m in kg, moment of inertia I in $\text{kg}\cdot\text{m}^2$ and torque τ in $\text{N}\cdot\text{m}$. A geometrically admissible leg or finger trajectory may demand unavailable torque.

One joint: geometry, energy and equation

Example pendulum: a point mass $m=0.5$ kg at the end of a massless rod $l=0.2$ m; the pivot is fixed. In the x-y plane, y points upward and q is measured from the downward vertical towards +x. Then $p=(l \sin q; -l \cos q)$, $v=\dot{q}$ in rad/s, $I=ml^2=0.02$ $\text{kg}\cdot\text{m}^2$. There is no friction or contact; positive τ_{act} increases q .

Newton and Euler: the torque of weight about the pivot is $-mgl \sin q$, so $I\ddot{q}=\tau_{\text{act}}-mgl \sin q$.

Lagrange: $K=lv^2/2$, $U=mgl(1-\cos q)$, $L=K-U$. Substitution into $d/dt(\partial L/\partial v)-\partial L/\partial q=\tau_{\text{act}}$ gives the same equation, $I\ddot{q}+mgl \sin q=\tau_{\text{act}}$. These are two descriptions of the same mechanics; their agreement helps check the signs.

At $q=\pi/6$, an acceleration $\ddot{q}=1$ rad/s^2 requires, by inverse dynamics, $\tau_{\text{act}}=0.02\cdot 1+0.5\cdot 9.81\cdot 0.2\cdot 0.5=0.5105$ $\text{N}\cdot\text{m}$. Of this, 0.4905 $\text{N}\cdot\text{m}$ compensates for weight and 0.0200 $\text{N}\cdot\text{m}$ accelerates the mass. Forward dynamics calculates $\ddot{q}=(\tau_{\text{act}}-mgl \sin q)/I$. With $\tau_{\text{act}}=0$ at this position, $\ddot{q}=-24.525$ rad/s^2 : the link accelerates towards the downward vertical, even when starting from rest.

From a pendulum to two links

For a fixed-base chain of revolute joints: $M(q)\dot{v}+h(q,v)+g(q)=\tau_{\text{act}}+J^T f_{\text{ext}}$. Here $v=\dot{q}$; M relates accelerations to torques, h contains velocity effects, and g contains gravitational torques; f_{ext} is the force exerted by the environment on the endpoint, expressed in the same basis as the positional Jacobian J . Geometry determines centre-of-mass velocities, from which energy and M are calculated. At rest, $\tau_{\text{act}}=g-J^T f_{\text{ext}}$; motion adds inertial and velocity-dependent terms.

An actuator with a position interface does not become a source of known torque: use the validated M7 model. A moving body, distributed contacts and spatial tensors require a more detailed model; this module requires a clear pendulum calculation and the specified 2R dynamics.

Check torque, acceleration and energy calculations

In tasks 3-6, save both derivations and the numerical example. Independently estimate dU/dq using a central difference: at $q=\pi/6$, expect 0.4905 $\text{N}\cdot\text{m}$. Apply 0.5105 $\text{N}\cdot\text{m}$ to forward dynamics and recover 1 rad/s^2 . Then, without actuation or friction, compare energy $E=K+U$ at two integrator step sizes; reducing the step should reduce numerical drift. Agreement between inverse and forward dynamics does not rule out an error shared by both functions.

Included in the 75 hours: 1 hour from 9-10 for the two derivations; 1 hour from 15-16 for the pendulum; 1 hour from 33-34 for the forward and inverse dynamics check. This is a required reference example within the existing tasks. Calculations run on the CPU; no additional equipment, software or hours are required.

References: Tedrake, [The Simple Pendulum: equation of motion and the Undamped Pendulum section](#); Lynch/Park, [Modern Robotics, §8.1.1 and §8.1.4: Lagrange, Newton and Euler methods](#).